

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

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REC-CIS

Finish review

Question 1
Correct
Marked out of 1.00
Flag question

A binary number is a combination of 1s and 0s. Its n^{th} least significant digit is the n^{th} digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the 4^{th} least significant digit.

Example

number = 23

- Convert the decimal number 23 to binary number: $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.
- The value of the 4^{th} index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \leq \text{number} \leq 231$

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reset answer

```
1 /*  
2  * Complete the 'fourthBit' function below.  
3  *  
4  * The function is expected to return an INTEGER.  
5  * The function accepts INTEGER number as parameter.  
6  */  
7  
8 int fourthBit(int number)  
9 {  
10     int binary[32];  
11     int i=0;  
12     while (number>0)  
13     {  
14         binary[i]=number%2;  
15         number/=2;  
16         i++;  
17     }  
18     if (i>=4)  
19     {  
20         return binary[3];  
21     }  
22     else  
23         return 0;  
24 }
```

	Test	Expected	Got	
✓	printf("%d", fourthBit(32))	0	0	✓
✓	printf("%d", fourthBit(77))	1	1	✓

Passed all tests! ✓

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Question
Correct
Marked out of 1.00
Flag question

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the p^{th} element of the list, sorted ascending. If there is no p^{th} element, return 0.

Example

$n = 20$
 $p = 3$

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if $p = 3$, then 4 is returned. If $p > 6$, 0 would be returned.

Function Description

Complete the function `pthFactor` in the editor below.

`pthFactor` has the following parameter(s):

- `int n`: the integer whose factors are to be found
- `int p`: the index of the factor to be returned

Returns:

- `int`: the long integer value of the p^{th} integer factor of `n` or, if there is no factor at that index, then 0 is returned

Constraints

REC-CIS

```
2  * Complete the 'pthFactor' function below.
3  *
4  * The function is expected to return a LONG_INTEGER.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER n
7  * 2. LONG_INTEGER p
8  */
9
10 long pthFactor(long n, long p)
11 {
12     int count=0;
13     for(long i=1;i<=n;i++)
14     {
15         if (n%i==0)
16         {
17             count++;
18             if(count==p)
19             {
20                 return i;
21             }
22         }
23     }
24     return 0;
25 }
```

Test	Expected	Got
✓ printf("%ld", pthFactor(10, 3))	5	5 ✓
✓ printf("%ld", pthFactor(10, 5))	0	0 ✓
✓ printf("%ld", pthFactor(1, 1))	1	1 ✓

Passed all tests! ✓